

# Uniform-shift CML for DIF

## Validation of the prototype estimator behind the planned RMdifImpact()

Magnus Johansson

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### Purpose

The planned DIF impact function (`dev/dif-impact-design.md`, Part B) needs a split model in which each DIF item has its own location in each group while the remaining items anchor the scale. `psychotools::pcmodel()` cannot fit an item-split dataset (see `dev/psychotools-na-bug/`), and a free split has two further drawbacks. It adds one parameter per threshold instead of one per item, and it fails when a category is empty in one group.

The prototype in `dev/dif_shift_cml.R` fits a **uniform-shift** model instead. Item thresholds are shared, and group 2 has every threshold of a DIF item shifted by  $d_i$ . With two groups the conditional likelihood is the sum of the two groups' PCM conditional likelihoods, so it is built from category counts and score distributions per group and the elementary symmetric functions from `psychotools::elementary_symmetric_functions()`.

This note checks four things before the estimator is used anywhere else:

1. With no DIF items it reproduces `pcmodel()`.
2. The analytic gradient matches numerical derivatives.
3. With one DIF item and a large sample it agrees with a free split (patched `pcmodel()` and `eRm::PCM()`), and it fits a category that is empty in one group, where the free split cannot.
4. Recovery: bias, SE calibration and CI coverage of  $\hat{d}$  by group size and item location. This part is a simulation and runs only with `params$run_recovery = TRUE`.

```
library(psychotools)
source("dif_shift_cml.R")
set.seed(20261003)

thr_items <- function(locs, steps) {
  out <- lapply(locs, function(l) l + steps)
  names(out) <- paste0("I", seq_along(locs))
}
```

```
out
}
```

## 1. No DIF items: agreement with `pcmodel()`

With an empty DIF set the shift model is the pooled PCM, and the group split does not change the conditional likelihood. The fit must match `pcmodel()` to optimiser tolerance.

```
v1_cases <- list(
  dich_k10_n300 = list(n = 300, thr = thr_items(seq(-2, 2, length.out = 10), 0)),
  poly3_k8_n500 = list(n = 500, thr = thr_items(seq(-1.5, 1.5, length.out = 8),
c(-0.5, 0.5))),
  poly5_k6_n200 = list(n = 200, thr = thr_items(seq(-1, 1, length.out = 6), c(-1.2,
-0.4, 0.4, 1.2))),
  mixed_k7_n400 = list(n = 400, thr = c(
    thr_items(c(-1, 0, 1), 0),
    setNames(thr_items(c(-0.5, 0.5), c(-0.6, 0.6)), c("I4", "I5")),
    setNames(thr_items(c(-0.8, 0.8), c(-1, 0, 1)), c("I6", "I7"))
  ))
)
v1 <- do.call(rbind, lapply(names(v1_cases), function(nm) {
  cs <- v1_cases[[nm]]
  s <- sim_dif(cs$n / 2, cs$n / 2, cs$thr)
  f <- shift_cml(s$X, s$g, hessian = FALSE)
  p <- pcmodel(s$X, nullcats = "downcode")
  data.frame(
    case = nm,
    loglik_diff = f$loglik - as.numeric(logLik(p)),
    max_par_diff = max(abs(f$psi[-1] - coef(p))),
    convergence = f$convergence
  )
}))
v1
```

| case          | loglik_diff | max_par_diff | convergence |
|---------------|-------------|--------------|-------------|
| dich_k10_n300 | 0           | 0e+00        | 0           |
| poly3_k8_n500 | 0           | 9e-07        | 0           |
| poly5_k6_n200 | 0           | 4e-07        | 0           |
| mixed_k7_n400 | 0           | 0e+00        | 0           |

## 2. Gradient check

Central differences with step  $10^{-6}$  at random parameter values, with two DIF items, for dichotomous and polytomous data.

```
grad_check <- function(s, dif) {
  prep <- .shift_prep(s$X, s$g, dif)
  par <- rnorm(prepare$P - 1L + length(prepare$dif_items), 0, 0.5)
  h <- 1e-6
  num <- vapply(seq_along(par), function(i) {
    e <- replace(numeric(length(par)), i, h)
    (.shift_negll(par + e, prep) - .shift_negll(par - e, prep)) / (2 * h)
  }, numeric(1))
}
```

```

    }, numeric(1))
    ana <- .shift_grad(par, prep)
    c(max_abs = max(abs(num - ana)), max_rel = max(abs(num - ana) / pmax(1, abs(ana))))
  }
  s_d <- sim_dif(300, 300, v1_cases$dich_k10_n300$thr, dif = c(I3 = 0.5))
  s_p <- sim_dif(300, 300, v1_cases$poly5_k6_n200$thr, dif = c(I2 = 0.5))
  rbind(
    dich = grad_check(s_d, c("I3", "I7")),
    poly = grad_check(s_p, c("I2", "I5"))
  )

```

```

      max_abs      max_rel
dich 3.475298e-07 5.930250e-09
poly 6.859418e-07 1.206553e-07

```

### 3. Agreement with a free split, and the sparse case

#### Large sample, uniform DIF

Eight items with three categories,  $n = 2000$  per group, group 2 0.5 SD higher,  $d = 0.8$  on I4. The free split fits I4 as two columns. Its mean threshold difference should agree closely with  $\hat{d}$ , and its log-likelihood can only be higher, since the shift model is nested in it.

```

thr8 <- thr_items(seq(-1.5, 1.5, length.out = 8), c(-0.5, 0.5))
s <- sim_dif(2000, 2000, thr8, dif = c(I4 = 0.8), mu2 = 0.5)
f_shift <- shift_cml(s$X, s$g, "I4")
sp <- split_items(s$X, s$g, "I4")
f_free <- pcmodel_fixed(sp)
tp <- threshpar(f_free)
e_free <- eRm::PCM(sp)
te <- eRm::thresholds(e_free)$threshtable[[1]]
lr <- 2 * (as.numeric(logLik(f_free)) - f_shift$loglik)
data.frame(
  estimate = c("shift d-hat", "free split, pcmodel (patched)", "free split, eRm"),
  d = c(f_shift$d, mean(tp$I4_g2 - tp$I4_g1), mean(te["I4_g2", -1] - te["I4_g1", -1])),
  se = c(f_shift$se_d, NA, NA),
  loglik = c(f_shift$loglik, as.numeric(logLik(f_free)), as.numeric(e_free$loglik))
)

```

| estimate                      | d         | se        | loglik    |
|-------------------------------|-----------|-----------|-----------|
| shift d-hat                   | 0.7458902 | 0.0524789 | -18126.47 |
| free split, pcmodel (patched) | 0.7450464 | NA        | -18126.13 |
| free split, eRm               | 0.7450479 | NA        | -18126.13 |

```

c(LR_free_vs_shift = lr, df = 1, p = pchisq(lr, df = 1, lower.tail = FALSE))

```

```

LR_free_vs_shift      df      p
0.6789925      1.0000000 0.4099339

```

The eRm log-likelihood is on a different constant from psychotools and is shown for completeness only. The LR statistic compares the two psychotools fits. Under a uniform-shift generating model it should be unremarkable.

## Large sample, threshold-specific DIF

The same design with only the second threshold of I4 shifted (by 0.8). Now the shift model is wrong for I4, the free split is right, and the LR test should detect it. This is the DSF case left for a later version, and this check shows what the shift model averages over.

```
thr_dsf <- thr8
s2 <- sim_dif(2000, 2000, thr8, mu2 = 0.5)
# Rebuild group 2 for I4 with only the upper threshold shifted.
g2 <- s2$g == "g2"
s2$X[g2, "I4"] <- sim_pcm(s2$theta[g2], list(I4 = thr8$I4 + c(0, 0.8)))
f_shift2 <- shift_cml(s2$X, s2$g, "I4")
f_free2 <- pmodel_fixed(split_items(s2$X, s2$g, "I4"))
tp2 <- threshpar(f_free2)
lr2 <- 2 * (as.numeric(logLik(f_free2)) - f_shift2$loglik)
list(
  shift_d = f_shift2$d,
  free_threshold_diffs = tp2$I4_g2 - tp2$I4_g1,
  LR = c(stat = lr2, p = pchisq(lr2, df = 1, lower.tail = FALSE))
)
```

```
$shift_d
      I4
0.3845857

$free_threshold_diffs
      C1      C2
0.006439281 0.702362430

$LR
      stat      p
2.667301e+01 2.409639e-07
```

## A category empty in one group

A small focal group ( $n = 60$ ) on a DIF item with four categories, where the top category is never used in group 2. Here the emptiness is forced by recoding group 2's top responses down one step, to make the demonstration deterministic. The shift model should still return a finite  $\hat{d}$  with an SE, while the free split has no information on the top threshold in group 2.

```
thr_sp <- c(thr_items(seq(-1, 1, length.out = 7), c(-0.5, 0.5)),
            list(I8 = 0.5 + c(-1, 0, 1)))
s3 <- sim_dif(300, 60, thr_sp, dif = c(I8 = 0.4), mu2 = 0)
g2 <- s3$g == "g2"
s3$X[g2 & s3$X[, "I8"] == 3, "I8"] <- 2L
table(group = s3$g, I8 = s3$X[, "I8"])
```

```
      I8
group  0  1  2  3
```

```
g1 71 105 73 51
g2 18 23 19 0
```

```
f_sp <- shift_cml(s3$X, s3$g, "I8")
c(d = f_sp$d, se = f_sp$se_d, convergence = f_sp$convergence)
```

```
      d.I8      se.I8 convergence
0.3494164 0.2014551 0.0000000
```

```
sp3 <- split_items(s3$X, s3$g, "I8")
free_pc <- tryCatch(threshpar(pcmmodel_fixed(sp3))[c("I8_g1", "I8_g2")],
                    error = function(e) conditionMessage(e), warning = function(w)
conditionMessage(w))
free_erm <- tryCatch({
  te3 <- eRm::thresholds(eRm::PCM(sp3))$threshtable[[1]]
  te3[c("I8_g1", "I8_g2"), ]
}, error = function(e) conditionMessage(e))
list(pcmmodel_free_split = free_pc, eRm_free_split = free_erm)
```

```
$pcmmodel_free_split
$pcmmodel_free_split$I8_g1
      C1      C2      C3
-0.8626495 0.4939769 1.1157964
```

```
$pcmmodel_free_split$I8_g2
      C1      C2
-0.5657943 0.2958567
```

```
$eRm_free_split
      Location Threshold 1 Threshold 2 Threshold 3
I8_g1 0.43114081 -0.6805478 0.6760756 1.297895
I8_g2 0.04712612 -0.3837017 0.4779540 NA
```

## 4. Recovery

```
rec_thr <- thr_items(seq(-2, 2, length.out = 8), c(-0.6, 0.6))
rec_grid <- expand.grid(
  sizes = c("150/150", "400/400", "600/150", "100/50"),
  item = c("I5", "I8"), # I5 near the centre, I8 at +2 logits
  d = c(0, 0.5),
  stringsAsFactors = FALSE
)
rec_reps <- 500
rec_file <- "dif_shift_cml_recovery.rds"
```

The design crosses four group-size splits (balanced and unbalanced), a DIF item near the centre of the person distribution (I5, location 0.29) or at +2 logits (I8), and  $d \in \{0, 0.5\}$ . Group 1 has  $\theta \sim N(0, 1)$ , group 2  $\theta \sim N(0.5, 1)$ . Each of the 16 cells has 500 replications. Outcomes: bias and empirical SD of  $\hat{d}$ , mean model SE and its ratio to the empirical SD, 95% Wald coverage, the share of replications where a category of the DIF item is empty in group 2, and failures (no convergence or no finite SE).

```

one_rep <- function(cell, seed) {
  set.seed(seed)
  n <- as.integer(strsplit(cell$sizes, "/")[[1]])
  dif <- setNames(cell$d, cell$item)
  s <- sim_dif(n[1], n[2], rec_thr, dif = dif, mu2 = 0.5)
  empty <- any(tabulate(s[X[s$g == "g2", cell$item] + 1L, nbins = 3L) == 0L)
  f <- tryCatch(shift_cml(s[X, s$g, cell$item]), error = function(e) NULL)
  if (is.null(f)) return(c(d_hat = NA, se = NA, ok = 0, empty = empty))
  c(d_hat = unname(f$d), se = unname(f$se_d),
    ok = as.numeric(f$convergence == 0 && is.finite(f$se_d)), empty = empty)
}
t0 <- Sys.time()
rec <- do.call(rbind, lapply(seq_len(nrow(rec_grid)), function(i) {
  cell <- rec_grid[i, ]
  res <- parallel::mclapply(seq_len(rec_reps), function(r) one_rep(cell, 1e5 * i
+ r),
                          mc.cores = 10L)
  data.frame(cell, do.call(rbind, res))
}))
attr(rec, "runtime") <- Sys.time() - t0
saveRDS(rec, rec_file)

```

```

if (file.exists(rec_file)) {
  rec <- readRDS(rec_file)
  agg <- do.call(rbind, lapply(split(rec, rec[c("sizes", "item", "d")], drop = TRUE),
function(x) {
  y <- x[x$ok == 1, ]
  data.frame(
    sizes = x$sizes[1], item = x$item[1], d = x$d[1],
    bias = mean(y$d_hat - y$d),
    emp_sd = sd(y$d_hat),
    mean_se = mean(y$se),
    se_ratio = mean(y$se) / sd(y$d_hat),
    coverage = mean(abs(y$d_hat - y$d) <= 1.96 * y$se),
    empty_cat = mean(x$empty),
    failed = mean(x$ok != 1)
  )
}))
agg <- agg[order(agg$item, agg$d, agg$sizes), ]
rownames(agg) <- NULL
num <- c("bias", "emp_sd", "mean_se", "se_ratio", "coverage", "empty_cat", "failed")
agg[num] <- lapply(agg[num], round, 3)
names(agg)[names(agg) == "empty_cat"] <- "empty"
agg
} else {
  cat("Recovery not run. Run with -P run_recovery:true to run it.\n")
}

```

| sizes   | item | d   | bias   | emp_sd | mean_se | se_ratio | coverage | empty | failed |
|---------|------|-----|--------|--------|---------|----------|----------|-------|--------|
| 100/50  | I5   | 0.0 | 0.002  | 0.297  | 0.297   | 1.002    | 0.958    | 0.000 | 0      |
| 150/150 | I5   | 0.0 | -0.004 | 0.203  | 0.197   | 0.971    | 0.950    | 0.000 | 0      |
| 400/400 | I5   | 0.0 | 0.006  | 0.124  | 0.120   | 0.967    | 0.928    | 0.000 | 0      |

| sizes   | item | d   | bias   | emp_sd | mean_se | se_ratio | coverage | empty | failed |
|---------|------|-----|--------|--------|---------|----------|----------|-------|--------|
| 600/150 | I5   | 0.0 | −0.001 | 0.163  | 0.155   | 0.951    | 0.948    | 0.000 | 0      |
| 100/50  | I5   | 0.5 | −0.004 | 0.287  | 0.298   | 1.038    | 0.968    | 0.000 | 0      |
| 150/150 | I5   | 0.5 | 0.011  | 0.191  | 0.197   | 1.033    | 0.966    | 0.000 | 0      |
| 400/400 | I5   | 0.5 | 0.009  | 0.123  | 0.120   | 0.976    | 0.946    | 0.000 | 0      |
| 600/150 | I5   | 0.5 | 0.002  | 0.156  | 0.155   | 0.993    | 0.954    | 0.000 | 0      |
| 100/50  | I8   | 0.0 | 0.020  | 0.389  | 0.372   | 0.956    | 0.936    | 0.036 | 0      |
| 150/150 | I8   | 0.0 | −0.002 | 0.254  | 0.249   | 0.981    | 0.950    | 0.000 | 0      |
| 400/400 | I8   | 0.0 | 0.003  | 0.138  | 0.152   | 1.104    | 0.968    | 0.000 | 0      |
| 600/150 | I8   | 0.0 | 0.002  | 0.197  | 0.190   | 0.965    | 0.950    | 0.000 | 0      |
| 100/50  | I8   | 0.5 | 0.050  | 0.408  | 0.407   | 0.999    | 0.958    | 0.124 | 0      |
| 150/150 | I8   | 0.5 | −0.004 | 0.275  | 0.265   | 0.962    | 0.950    | 0.000 | 0      |
| 400/400 | I8   | 0.5 | 0.008  | 0.158  | 0.161   | 1.023    | 0.968    | 0.000 | 0      |
| 600/150 | I8   | 0.5 | 0.024  | 0.227  | 0.211   | 0.932    | 0.934    | 0.002 | 0      |

## Findings

Recovery run 2026-10-03, 16 cells  $\times$  500 replications, 13 seconds on 10 cores. The Monte Carlo SE of a coverage rate near .95 is about .010, and of the bias about SD/22.

1. **The estimator reproduces `pcmodel()` and its gradient is right.** With no DIF items the largest parameter difference is below  $10^{-6}$  in all four cases, and the analytic gradient agrees with central differences to below  $10^{-6}$ .
2. **It agrees with a free split when DIF is uniform.** At 2000 per group,  $\hat{d} = 0.746$  against 0.745 from both the patched `pcmodel()` and eRm free splits, and the LR test of the shift model against the free split is unremarkable. When only one threshold carries the DIF, the LR test detects it, so the shift model's adequacy can be checked against a free split whenever the free split is estimable.
3. **It handles a category that is empty in one group.** It returns a finite  $\hat{d}$  with an SE, while eRm returns NA for that threshold and the patched `pcmodel()` drops it without notice. In the recovery run a category of the off-target item was empty in the focal group in 12.4% of replications at 100/50 with  $d = 0.5$ , with no failures.
4. **Recovery is good down to 100/50.** No replication failed. Bias is within 0.012 of zero for the central item. For the off-target item it reaches 0.050 at 100/50 and 0.024 at 600/150 with  $d = 0.5$ , about 2.5 Monte Carlo SEs, a small upward bias where the focal group is small and the item is far from it. Model SEs track the empirical SD (ratio 0.93 to 1.10), and Wald coverage is .928 to .968, within about 2.3 Monte Carlo SEs of .95 everywhere.

The prototype is fit for use in the impact study. Two things to carry forward: the small-sample bias off target is the same direction as the noise-floor question (study item 3), and the SEs are good enough that a Wald interval for  $\hat{d}$  can be reported directly.